

a* (carotenoid proxy) under metal stress: stratified analyses

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a* (carotenoid proxy) under metal stress: stratified analyses

Generated 2026-10-08. All numbers come from `report/tables/*.csv`; scripts are in `scripts/` and `scripts/strat/`; the data-quality caveats are in `analysis/heavy_metal_data_problems/HEAVY_METAL_DATA_PROBLEMS.md`.

Question. When *Rhodotorula* strains grow on plates with a heavy metal at increasing doses, does CIELAB a* (red-green axis, used here as a carotenoid proxy) rise because of stress, or do strains differ inherently in a*? Colony size confounds this, so every dose effect is shown with and without size.

Summary

1. **There is no general “stress raises a*”.** The effect depends on the metal. Over the full dose range, at fixed colony size: Cr -8.4, Cu -13.4, Pb -6.1 (a* units), Fe +3.8. Zn (-3.7, $p = 0.44$) is not informative (7 plates, one per dose).
2. **The dose-response is not linear.** Dose-as-factor models show Cr and Pb a* *rise* at low to middle doses (Cr up to +3.1 at dose 0.4; Pb up to about +7.5 at doses 10-15, at fixed size) and fall at the highest doses. Cu a* falls from the first doses, even where colonies are not smaller than control (area ratio 1.0-1.2 at doses 5-15). Fe rises steadily.
3. **Top-dose a* in Cr, Cu and Pb looks like a floor, and small colonies are a concern in Pb and Cr.** At the top dose a* collapses to a narrow band (about 5-6) for every strain, whatever its baseline. In Pb the smallest colonies (below about 1,850 px) have a* of 6-7 with SD 2-2.7 even at dose 0. Dropping colonies below 1,000-5,000 px (section 10) leaves Cu and Fe unchanged. The Cr top-dose drop persists in larger colonies (-9.1 with no filter, -10.7 at 5,000 px). The Pb rise at doses 5-15 persists (+4.9 to +9.5 at 2,000 px), but almost no Pb colonies at doses 20-30 pass the larger thresholds (242-265 wells at 2,000 px, 8-17 at 5,000 px), so the Pb inhibitory range cannot be tested.
4. **Genetic structure matters more than species labels.** *R. mucilaginosa* populations differ in baseline a* in all four metals (omnibus $p < 0.001$; population 5 is lowest, -2.6 to -4.0 a* units vs population 1). Species differences are metal-specific. *R. toruloides*, *R. dairenensis* and *R. diobovata* have lower estimates than *R. mucilaginosa* in all four metals, but each is individually significant in only one to three of them.
5. **Phylogenetic signal is strong.** Pagel’s lambda for baseline a* is 0.90-0.98 in Cr, Cu and Pb, also within *R. mucilaginosa* alone. Fe is lower (0.51; 0.23 size-adjusted).
6. **Regression to the mean inflates the baseline-vs-induction pattern.** Selecting the high-baseline third of strains on one half of the replicate wells and measuring the change on the other half gives a smaller gap than the naive analysis: Cu -1.7 vs -5.3, Cr -5.5 vs -6.9, Pb -4.2 vs -6.7. Fe is unaffected (+1.9 vs +1.6).

7. **Strain-specific change in a^* is only partly repeatable.** Correlation of the change between replicate halves: Fe 0.70, Cr 0.54, Pb 0.31, Cu 0.13.
8. **Run (batch) is a small share of variance (0-2%).** Only Cr shows heterogeneity in the dose effect between runs ($I^2 = 0.68$, $Q p = 0.025$): the effect weakens from -12.6 (run d000320) to -5.6 (run d000323). Each run is a different strain subset, so run and strain set cannot be separated.

1. Data and method (short)

- Source: DuckDB table `heavy_metal_measurement` (517,371 colony rows; 5 metals). Named strains only (502,051 rows). Unit of analysis is the **well**: the largest object per well and image, median over an absolute late window [T-24 h, T] after plate start, T = median plate imaging span per metal (Cr 114, Cu 114, Fe 89.9, Pb 107.5, Zn 107.6 h); wells need at least 2 images in the window.
- Models (lme4, REML): strain, run and plate are random effects. `dose_s` = concentration / maximum concentration. “At fixed size” adds $\ln(\text{area})$ centred on the control mean.
- Dose units are not given in the source. Cr spans 0-1.2 and the other metals 0-30, so metals are never pooled.
- Zinc: the two runs are treated as one experiment. It has 7 usable plates, one per dose, 1 well per strain per dose, and two disjoint strain sets (doses 0-15 and 10-30). Dose and plate cannot be separated, so Zinc is shown for completeness only and is not used in the stratified models.

2. Overall dose response

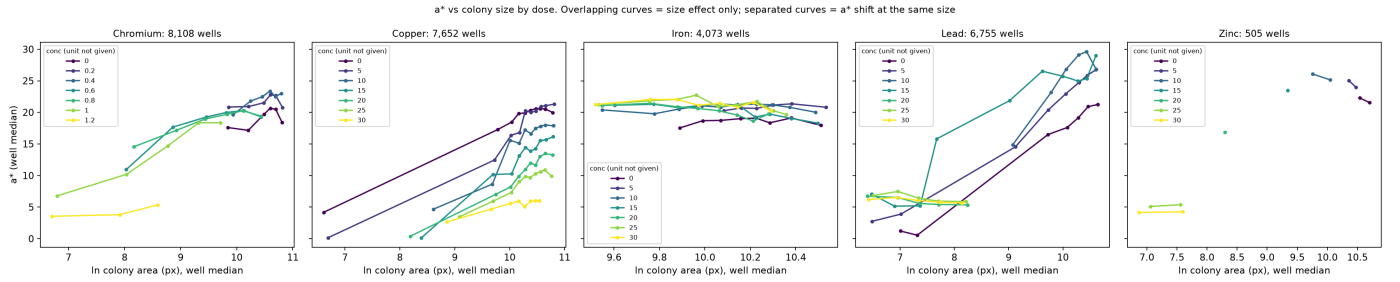


Figure 1. a^* against colony size by dose (well medians, binned). Overlapping curves mean a size effect only; separated curves mean a shift at the same size.

Metal	model	n_wells	n_strains	dose effect (0 to top)	SE	p	strain share @0	strain share @top	singular
Chromium	total	8108	316	-13.54	1.07	<0.001	0.50	0.24	False
Chromium	at fixed size	8108	316	-8.36	0.91	<0.001	0.55	0.24	False
Copper	total	7652	319	-15.86	0.39	<0.001	0.47	0.36	False
Copper	at fixed size	7652	319	-13.37	0.31	<0.001	0.51	0.28	False
Iron	total	4073	220	2.39	0.36	<0.001	0.80	0.89	False
Iron	at fixed size	4073	220	3.83	0.35	<0.001	0.82	0.91	False
Lead	total	6755	312	-16.77	1.54	<0.001	0.51	0.00	True
Lead	at fixed size	6755	312	-6.10	1.15	<0.001	0.49	0.03	False
Zinc	total	505	151	-16.93	4.49	0.01	0.26	0.72	False
Zinc	at fixed size	505	151	-3.66	4.46	0.44	0.41	0.66	False

Table 1. Dose effect on a^* from 0 to the top dose. “total” has no size term; “at fixed size” adds $\ln(\text{area})$. “strain share” is strain variance / (strain + plate + residual) at dose 0 and at the top dose. The Pb total model is a singular fit.

dose	Chromium	Copper	Iron	Lead	Zinc
0.20	2.17				
0.40	3.10				
0.60	1.18				
0.80	0.94				
1.00	-3.31				
1.20	-9.12				
5.00		0.05	2.53	4.44	5.43
10.00		-3.21	3.09	7.49	6.80
15.00		-5.56	2.87	7.56	6.91
20.00		-8.61	3.58	-0.90	5.38
25.00		-10.56	4.28	-0.65	1.37
30.00		-12.30	4.64	-1.41	-1.76

Table 2. a^* difference from 0 dose at fixed size (dose-as-factor model; SE 0.2-0.5 except Zn, about 8.8).

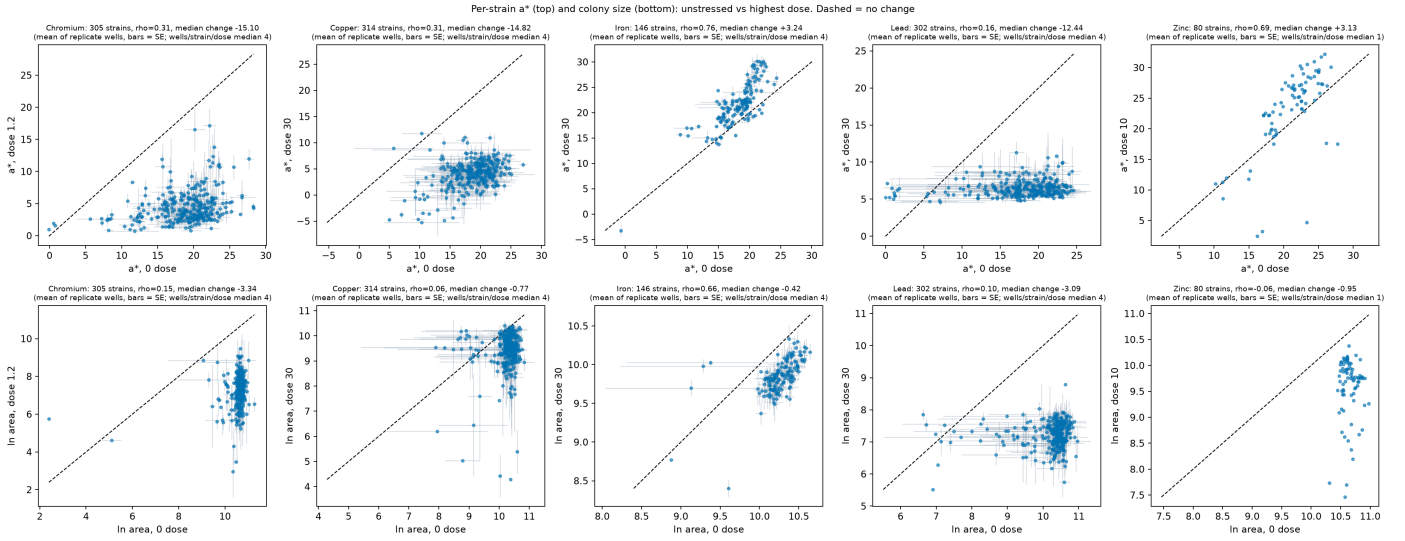


Figure 2. Per-strain a^* (top) and $\ln$ area (bottom), unstressed against top dose (mean of replicate wells, bars = SE). For Zinc the top dose shown is 15, the highest dose shared with dose 0.

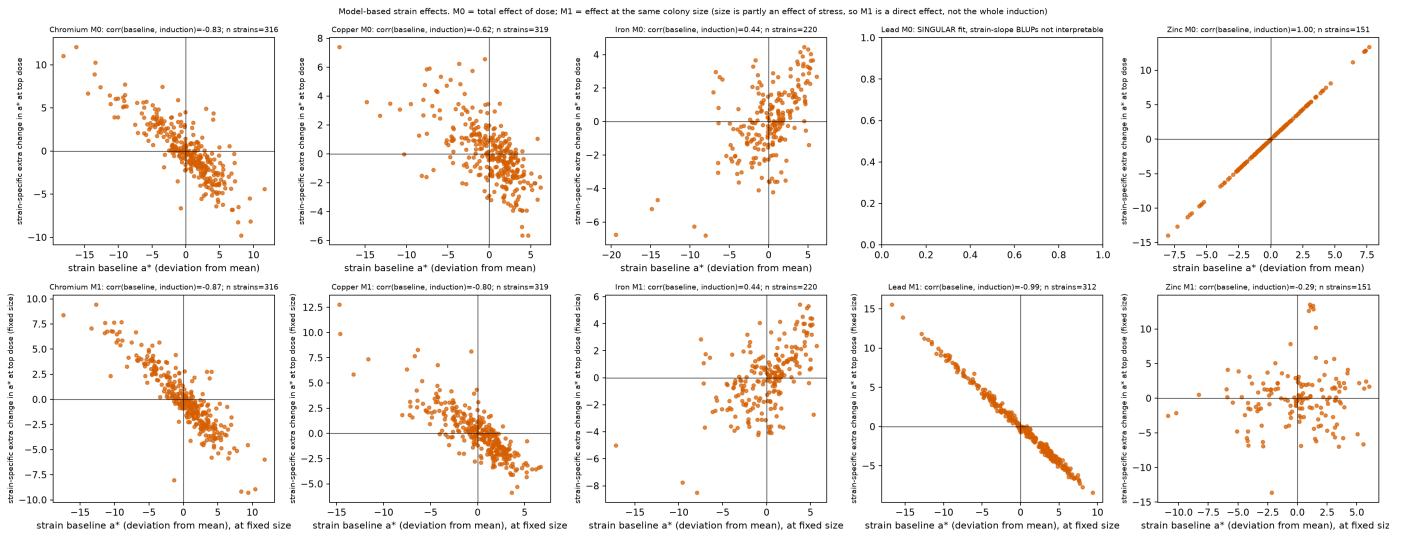


Figure 2b. Model-based strain effects: baseline a^* (x) against strain-specific extra change in a^* at the top dose (y), total (top row) and at fixed size (bottom row). Pb total is a singular fit and is not drawn. Pb at fixed size ($r = -0.99$) and Zn total ($r = +1.00$) lie on a straight line: the random-effect correlation is at the model boundary, so these points are a

rescaling of one random effect and carry no separate information.

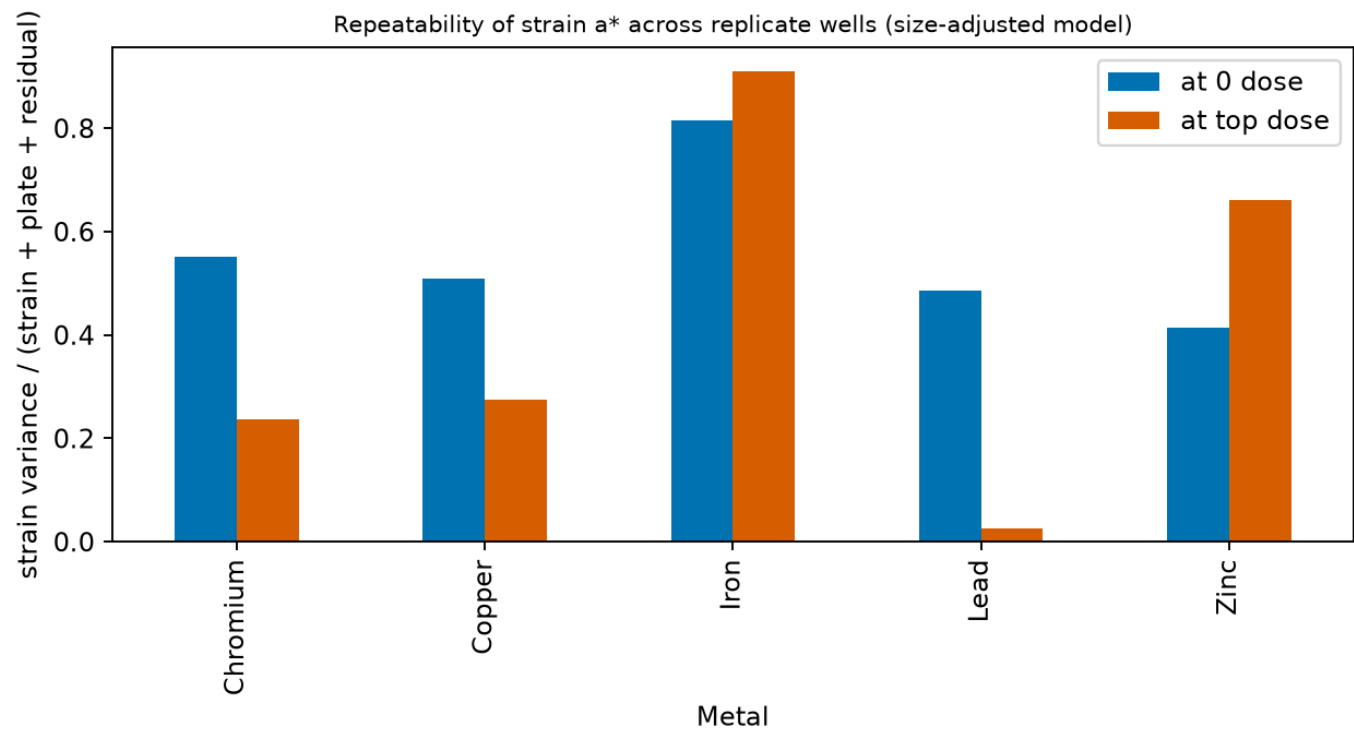


Figure 3. Share of a^* variance due to strain at 0 dose and at the top dose (size-adjusted model).

Reading. In Cr, Cu and Pb strains converge to a common low a^* at the top dose (strain share falls to 0.24, 0.28 and 0.03). Fe keeps its strain differences (0.82 to 0.91).

3. Which traits correlate with a^*

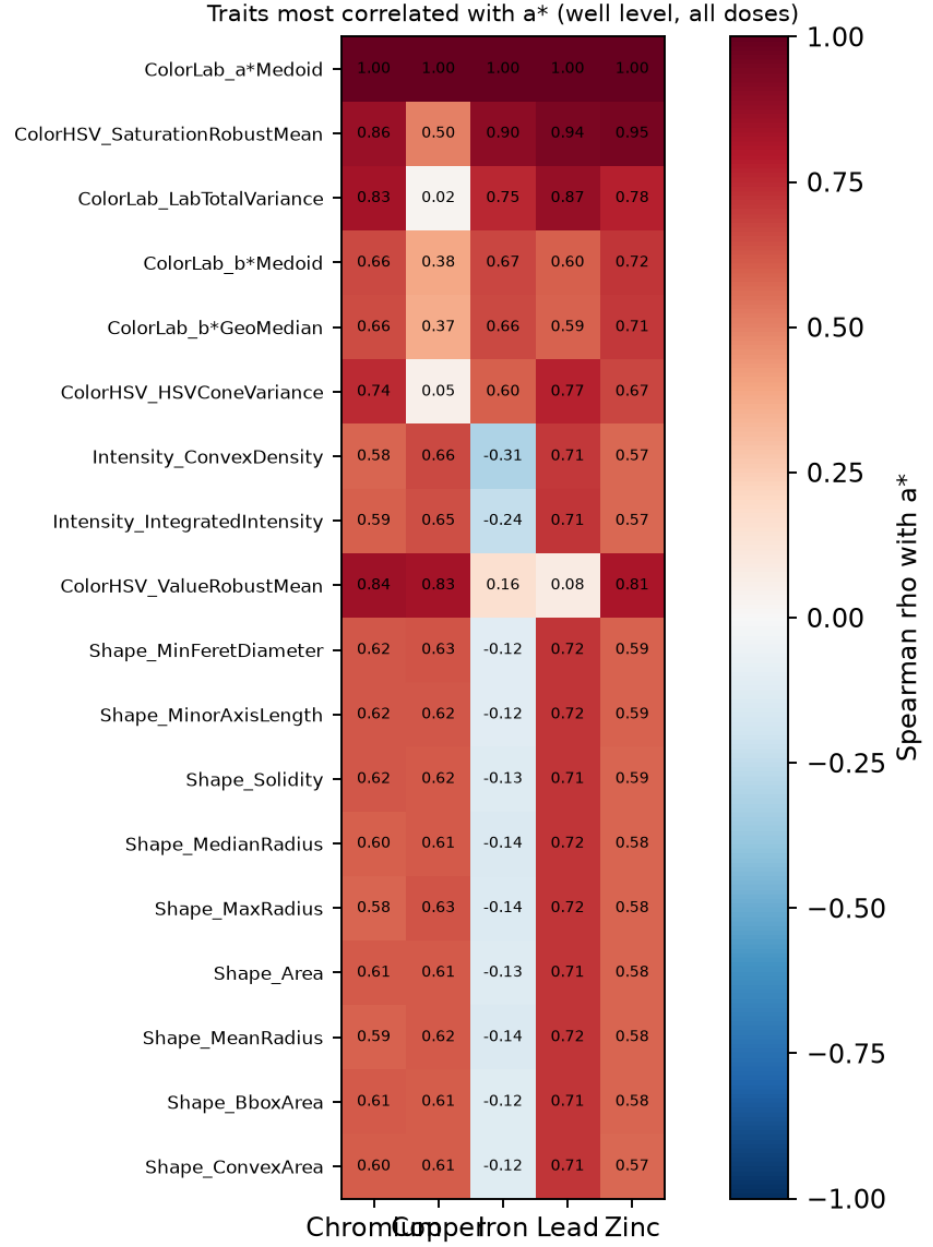


Figure 4. Spearman correlation of a^* with the traits most correlated with it (well level, all doses).

All doses (excluding ColorLab_a*Medoid, which is a^* itself):

trait	Chromium	Copper	Iron	Lead	Zinc	mean
ColorHSV_SaturationRobustMean	0.86	0.50	0.90	0.94	0.95	0.83
ColorLab_LabTotalVariance	0.83	0.02	0.75	0.87	0.78	0.65
ColorLab_b*Medoid	0.66	0.38	0.67	0.60	0.72	0.61
ColorLab_b*GeoMedian	0.66	0.37	0.66	0.59	0.71	0.60
ColorHSV_HSVConcVariance	0.74	0.05	0.60	0.77	0.67	0.57
Intensity_ConvexDensity	0.58	0.66	-0.31	0.71	0.57	0.57
Intensity_IntegratedIntensity	0.59	0.65	-0.24	0.71	0.57	0.55
ColorHSV_ValueRobustMean	0.84	0.83	0.16	0.08	0.81	0.55
Shape_MinFeretDiameter	0.62	0.63	-0.12	0.72	0.59	0.53
Shape_MinorAxisLength	0.62	0.62	-0.12	0.72	0.59	0.53

Control dose only:

trait	Chromium	Copper	Iron	Lead	Zinc	mean
ColorHSV_SaturationRobustMean	0.85	0.63	0.90	0.90	0.91	0.84
ColorLab_LabTotalVariance	0.79	0.74	0.85	0.81	0.93	0.83
ColorHSV_HSVConcVariance	0.68	0.62	0.79	0.74	0.84	0.73
ColorHSV_ValueRobustMean	0.69	0.55	0.57	0.70	0.47	0.60
ColorLab_b*Medoid	0.40	0.31	0.41	0.48	0.44	0.41
Intensity_StandardDeviationIntensity	-0.29	-0.56	-0.38	-0.42	-0.37	0.40
ColorLab_b*GeoMedian	0.39	0.30	0.40	0.47	0.42	0.40
Intensity_MedianIntensity	-0.36	-0.42	-0.38	-0.20	-0.55	0.38
ColorLab_DeltaE2000MeanFromMedoid	0.31	0.25	0.51	0.26	0.55	0.38
Texture_Correlation-deg090-scale05	-0.24	-0.46	-0.36	-0.30	-0.43	0.36

Within strain and dose (replicate wells only; Zinc has no replicates):

trait	Chromium	Copper	Iron	Lead	mean
ColorHSV_SaturationRobustMean	0.83	0.63	0.88	0.84	0.79
ColorHSV_ValueRobustMean	0.75	0.50	0.63	0.68	0.64
ColorLab_b*Medoid	0.67	0.45	0.73	0.60	0.61
ColorLab_b*GeoMedian	0.67	0.44	0.74	0.60	0.61
ColorLab_LabTotalVariance	0.67	0.27	0.55	0.69	0.55
ColorHSV_HSVConcVariance	0.58	0.21	0.44	0.63	0.47
Shape_MinorAxisLength	0.35	0.44	0.54	0.50	0.46
Shape_Solidity	0.35	0.44	0.54	0.50	0.46
Shape_MinFeretDiameter	0.34	0.44	0.54	0.50	0.46
Shape_MedianRadius	0.32	0.44	0.53	0.50	0.45

Reading. Saturation, b* and the colour-variance traits correlate most strongly with a*; these are partly the same colour information. Size traits (area, radius, Feret diameter, integrated intensity) correlate at 0.5-0.7 in Cr, Cu, Pb and Zn but with the opposite sign in Fe. Many size traits are near-duplicates of each other, so treat them as one family. Replicate wells within a strain and dose still show a size-a* correlation of about 0.3-0.54, so the link is not only between strains.

4. Species and population (stratification item 1)

Species is a fixed effect (reference *R. mucilaginosa*); strain, run and plate are random. Only species with at least 5 strains are included. Population labels exist for 201 *R. mucilaginosa* strains (file `analysis/gwas/data/prior_run_state/pop_assignment_at_`

Metal	model	F	p	p_BH
Chromium	baseline	7.74	<0.001	<0.001
Chromium	baseline_size_adjusted	6.40	<0.001	<0.001
Chromium	dose_response_size_adjusted	9.14	<0.001	<0.001
Chromium	population_baseline_size_adjusted	8.46	<0.001	<0.001
Chromium	population_dose_response_size_adjusted	14.18	<0.001	<0.001
Copper	baseline	1.66	0.12	0.13
Copper	baseline_size_adjusted	1.48	0.17	0.18
Copper	dose_response_size_adjusted	0.64	0.72	0.72
Copper	population_baseline_size_adjusted	5.46	<0.001	<0.001
Copper	population_dose_response_size_adjusted	6.56	<0.001	<0.001
Iron	baseline	4.40	<0.001	<0.001
Iron	baseline_size_adjusted	2.97	0.01	0.01
Iron	dose_response_size_adjusted	1.96	0.07	0.09
Iron	population_baseline_size_adjusted	18.96	<0.001	<0.001
Iron	population_dose_response_size_adjusted	7.68	<0.001	<0.001
Lead	baseline	15.76	<0.001	<0.001

Metal	model	F	p	p_BH
Lead	baseline_size_adjusted	4.33	<0.001	<0.001
Lead	dose_response_size_adjusted	8.75	<0.001	<0.001
Lead	population_baseline_size_adjusted	12.00	<0.001	<0.001
Lead	population_dose_response_size_adjusted	13.73	<0.001	<0.001

Table 3. Omnibus tests (*F* test; Benjamini-Hochberg across all rows).

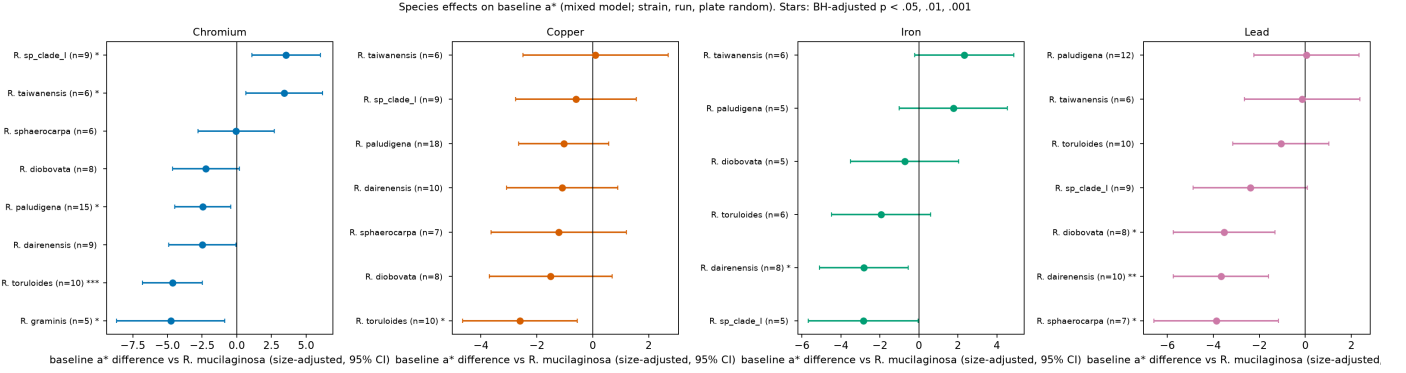


Figure 5. Species effect on baseline a^* relative to *R. mucilaginosa* (size-adjusted).

Metal	species	n_strains	estimate	se	p	p_BH
Chromium	<i>R. dairenensis</i>	9	-2.48	1.23	0.05	0.11
Chromium	<i>R. diobovata</i>	8	-2.22	1.23	0.07	0.14
Chromium	<i>R. graminis</i>	5	-4.76	1.99	0.02	0.05
Chromium	<i>R. paludigena</i>	15	-2.45	1.03	0.02	0.05
Chromium	<i>R. sp_clade_I</i>	9	3.56	1.26	0.01	0.03
Chromium	<i>R. spaeocarpha</i>	6	-0.04	1.41	0.98	0.98
Chromium	<i>R. taiwanensis</i>	6	3.42	1.41	0.02	0.05
Chromium	<i>R. toruloides</i>	10	-4.63	1.10	<0.001	<0.001
Copper	<i>R. dairenensis</i>	10	-1.09	1.02	0.28	0.40
Copper	<i>R. diobovata</i>	8	-1.50	1.12	0.18	0.30
Copper	<i>R. paludigena</i>	18	-1.04	0.82	0.21	0.31
Copper	<i>R. sp_clade_I</i>	9	-0.60	1.10	0.59	0.71
Copper	<i>R. spaeocarpha</i>	7	-1.22	1.23	0.32	0.41
Copper	<i>R. taiwanensis</i>	6	0.10	1.33	0.94	0.98
Copper	<i>R. toruloides</i>	10	-2.60	1.05	0.01	0.05
Iron	<i>R. dairenensis</i>	8	-2.83	1.16	0.02	0.05
Iron	<i>R. diobovata</i>	5	-0.73	1.41	0.61	0.71
Iron	<i>R. paludigena</i>	5	1.77	1.42	0.21	0.31
Iron	<i>R. sp_clade_I</i>	5	-2.85	1.44	0.05	0.11
Iron	<i>R. taiwanensis</i>	6	2.33	1.30	0.07	0.14
Iron	<i>R. toruloides</i>	6	-1.94	1.30	0.14	0.24
Lead	<i>R. dairenensis</i>	10	-3.67	1.05	<0.001	0.01
Lead	<i>R. diobovata</i>	8	-3.53	1.13	0.00	0.02
Lead	<i>R. paludigena</i>	12	0.04	1.17	0.97	0.98
Lead	<i>R. sp_clade_I</i>	9	-2.39	1.26	0.06	0.13
Lead	<i>R. spaeocarpha</i>	7	-3.87	1.38	0.01	0.03
Lead	<i>R. taiwanensis</i>	6	-0.14	1.28	0.91	0.98
Lead	<i>R. toruloides</i>	10	-1.06	1.06	0.32	0.41

Table 4. Species baseline contrasts (size-adjusted).

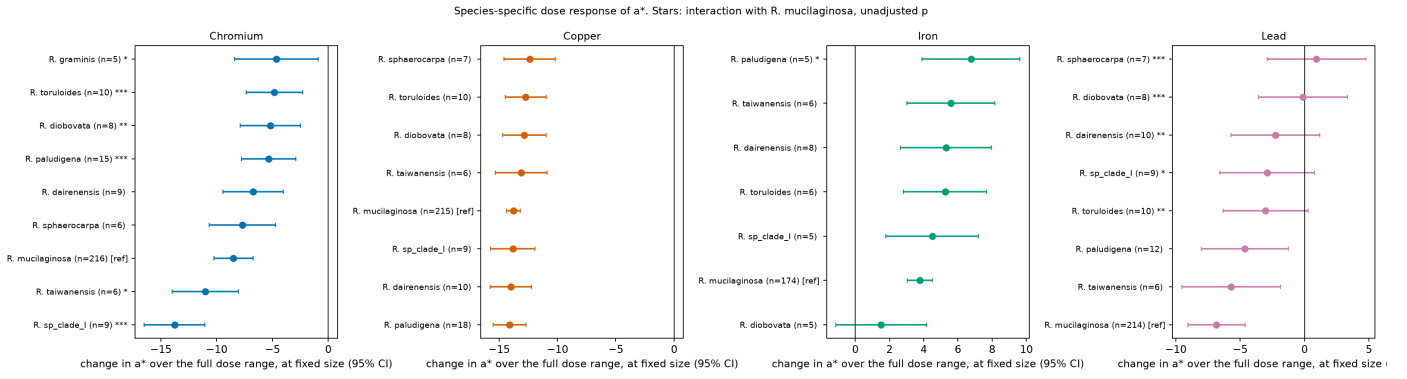


Figure 6. Species-specific change in a^* over the full dose range at fixed size.

Metal	species	n_strains	slope_full_range	se	p vs mucilaginosa
Chromium	<i>R. mucilaginosa</i>	216	-8.49	0.90	
Chromium	<i>R. dairenensis</i>	9	-6.74	1.37	0.10
Chromium	<i>R. diobovata</i>	8	-5.19	1.38	0.00
Chromium	<i>R. graminis</i>	5	-4.64	1.91	0.03
Chromium	<i>R. paludigena</i>	15	-5.34	1.24	<0.001
Chromium	<i>R. sp_clade_I</i>	9	-13.77	1.39	<0.001
Chromium	<i>R. sphaerocarpa</i>	6	-7.70	1.51	0.53
Chromium	<i>R. taiwanensis</i>	6	-11.02	1.52	0.04
Chromium	<i>R. toruloides</i>	10	-4.83	1.29	<0.001
Copper	<i>R. mucilaginosa</i>	215	-13.77	0.30	
Copper	<i>R. dairenensis</i>	10	-13.99	0.91	0.80
Copper	<i>R. diobovata</i>	8	-12.83	0.96	0.32
Copper	<i>R. paludigena</i>	18	-14.10	0.72	0.63
Copper	<i>R. sp_clade_I</i>	9	-13.82	0.96	0.95
Copper	<i>R. sphaerocarpa</i>	7	-12.38	1.12	0.21
Copper	<i>R. taiwanensis</i>	6	-13.10	1.13	0.55
Copper	<i>R. toruloides</i>	10	-12.73	0.89	0.24
Iron	<i>R. mucilaginosa</i>	174	3.77	0.37	
Iron	<i>R. dairenensis</i>	8	5.30	1.36	0.25
Iron	<i>R. diobovata</i>	5	1.52	1.35	0.10
Iron	<i>R. paludigena</i>	5	6.77	1.46	0.04
Iron	<i>R. sp_clade_I</i>	5	4.49	1.39	0.61
Iron	<i>R. taiwanensis</i>	6	5.59	1.31	0.17
Iron	<i>R. toruloides</i>	6	5.25	1.24	0.23
Lead	<i>R. mucilaginosa</i>	214	-6.83	1.14	
Lead	<i>R. dairenensis</i>	10	-2.26	1.75	0.00
Lead	<i>R. diobovata</i>	8	-0.11	1.76	<0.001
Lead	<i>R. paludigena</i>	12	-4.63	1.72	0.11
Lead	<i>R. sp_clade_I</i>	9	-2.90	1.87	0.01
Lead	<i>R. sphaerocarpa</i>	7	0.93	1.96	<0.001
Lead	<i>R. taiwanensis</i>	6	-5.71	1.96	0.49
Lead	<i>R. toruloides</i>	10	-3.02	1.69	0.00

Table 5. Species-specific dose slopes (a^* change over the full dose range).

Metal	n_other_strains	baseline_diff	baseline_se	baseline_p	slope_diff	slope_se	slope_p
Chromium	68	-1.38	0.53	0.01	1.29	0.48	0.01
Copper	68	-1.19	0.45	0.01	0.34	0.39	0.38
Iron	35	-0.84	0.60	0.17	1.02	0.60	0.09
Lead	62	-2.09	0.51	0.00	4.27	0.63	0.00

Table 6. *R. mucilaginosa* against all other species pooled (contrast only).

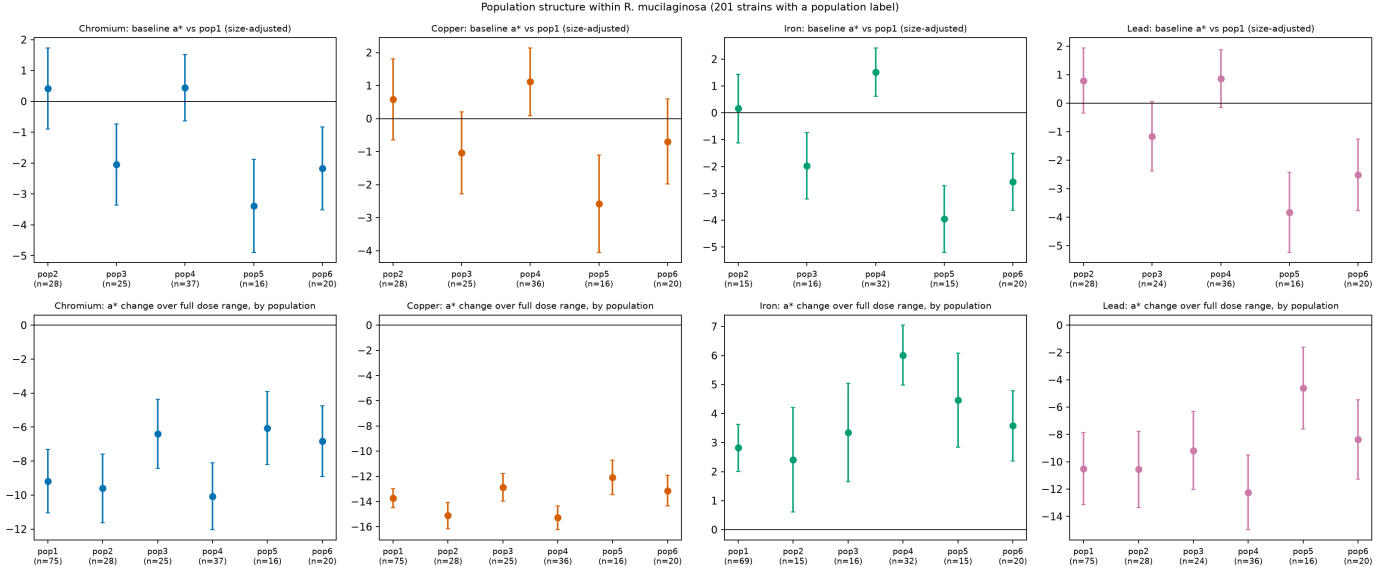


Figure 7. Populations within *R. mucilaginosa*: baseline a^* (top) and dose response (bottom).

Metal	population	n_strains	estimate	se	p
Chromium	pop2	28	0.42	0.67	0.53
Chromium	pop3	25	-2.05	0.67	0.00
Chromium	pop4	37	0.45	0.55	0.42
Chromium	pop5	16	-3.39	0.77	<0.001
Chromium	pop6	20	-2.17	0.68	0.00
Copper	pop2	28	0.58	0.62	0.35
Copper	pop3	25	-1.03	0.63	0.10
Copper	pop4	36	1.11	0.52	0.04
Copper	pop5	16	-2.58	0.75	<0.001
Copper	pop6	20	-0.69	0.66	0.29
Iron	pop2	15	0.16	0.65	0.81
Iron	pop3	16	-1.97	0.63	0.00
Iron	pop4	32	1.52	0.46	0.00
Iron	pop5	15	-3.96	0.63	<0.001
Iron	pop6	20	-2.57	0.54	<0.001
Lead	pop2	28	0.79	0.58	0.18
Lead	pop3	24	-1.16	0.62	0.06
Lead	pop4	36	0.86	0.52	0.10
Lead	pop5	16	-3.83	0.72	<0.001
Lead	pop6	20	-2.51	0.64	<0.001

Table 7. Population baseline contrasts against pop1 (size-adjusted).

Metal	population	n_strains	slope_full_range	se	z_p	vs_pop1_p
Chromium	pop1	75	-9.18	0.96	0.00	
Chromium	pop2	28	-9.61	1.03	0.00	0.48
Chromium	pop3	25	-6.39	1.03	0.00	0.00
Chromium	pop4	37	-10.07	1.00	0.00	0.07
Chromium	pop5	16	-6.06	1.09	0.00	0.00
Chromium	pop6	20	-6.82	1.06	0.00	0.00
Copper	pop1	75	-13.74	0.38	0.00	
Copper	pop2	28	-15.11	0.53	0.00	0.02

Metal	population	n_strains	slope_full_range	se	z_p	vs_pop1_p
Copper	pop3	25	-12.88	0.56	0.00	0.16
Copper	pop4	36	-15.29	0.48	0.00	0.00
Copper	pop5	16	-12.08	0.70	0.00	0.02
Copper	pop6	20	-13.14	0.62	0.00	0.34
Iron	pop1	69	2.82	0.42	0.00	
Iron	pop2	15	2.41	0.92	0.01	0.65
Iron	pop3	16	3.34	0.86	0.00	0.54
Iron	pop4	32	6.01	0.53	0.00	0.00
Iron	pop5	15	4.46	0.83	0.00	0.05
Iron	pop6	20	3.58	0.62	0.00	0.23
Lead	pop1	75	-10.50	1.34	0.00	
Lead	pop2	28	-10.56	1.43	0.00	0.94
Lead	pop3	24	-9.18	1.46	0.00	0.12
Lead	pop4	36	-12.24	1.39	0.00	0.01
Lead	pop5	16	-4.62	1.53	0.00	0.00
Lead	pop6	20	-8.36	1.48	0.00	0.01

Table 8. Population dose slopes.

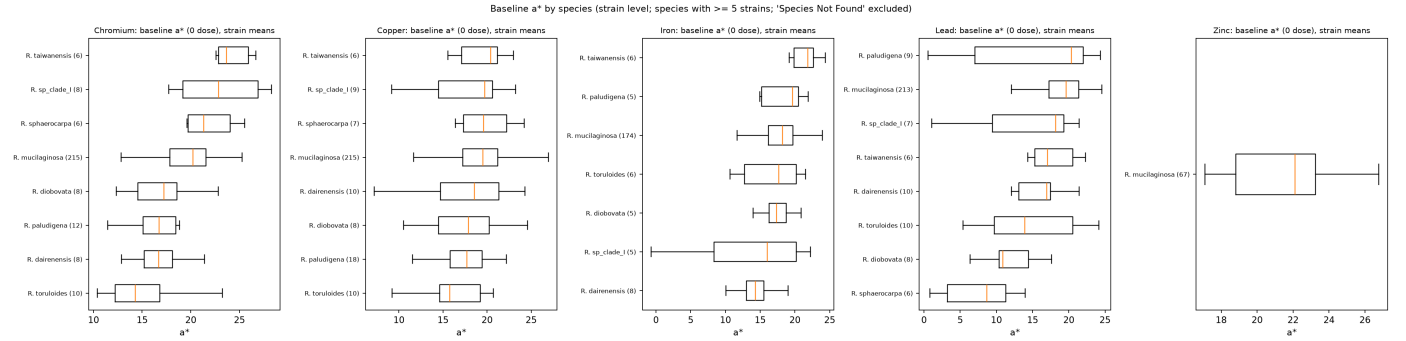


Figure 8. Baseline a^* by species (strain means; species with at least 5 strains).

Reading. - The population effect is the most consistent result in this section. Population 5 is lowest in all four metals (Cr -3.4, Cu -2.6, Fe -4.0, Pb -3.8). Population 6 is also low in Cr, Fe and Pb, and population 3 in Cr and Fe. Population 4 has the highest or tied-highest estimate in every metal, but differs from pop1 significantly only in Cu and Fe. The same pattern in four independent metal screens suggests a real genetic effect. - Species effects are inconsistent across metals (for example sp_clade_I is +3.6 in Cr and -2.4 in Pb), and the non-mucilaginosa species have 5-18 strains each, so single-species contrasts are noisy. Cu shows no species effect overall ($F p = 0.17$). - Pooled “other species” are lower than *R. mucilaginosa* in baseline a^* in all four metals (-0.8 to -2.1), and respond differently in Pb (slope difference +4.3).

5. Phylogenetic signal (item 2)

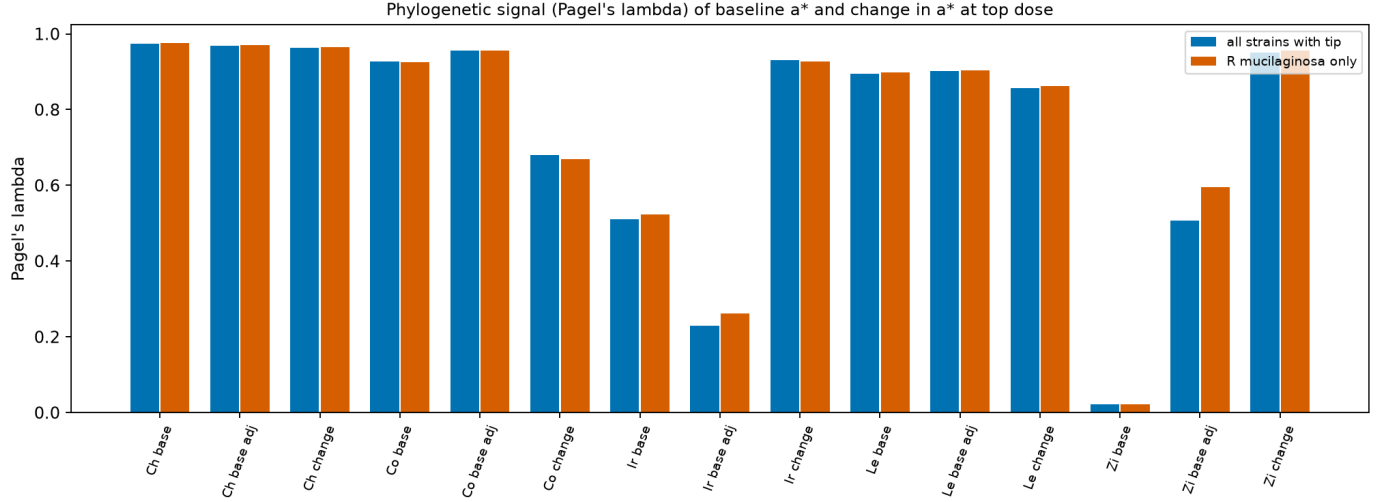


Figure 9. Pagel's lambda of baseline a^* and of the change in a^* at the top dose.

Metal	scope	trait	n_strains	pagel_lambda	p (lambda > 0)
Chromium	all_strains_with_tip	baseline a^*	256	0.97	<0.001
Chromium	all_strains_with_tip	baseline a^* (size-adjusted)	256	0.97	<0.001
Chromium	all_strains_with_tip	change in a^* at dose 1.2	254	0.96	<0.001
Chromium	R_mucilaginosa_only	baseline a^*	201	0.97	<0.001
Chromium	R_mucilaginosa_only	baseline a^* (size-adjusted)	201	0.97	<0.001
Chromium	R_mucilaginosa_only	change in a^* at dose 1.2	199	0.96	<0.001
Copper	all_strains_with_tip	baseline a^*	264	0.93	<0.001
Copper	all_strains_with_tip	baseline a^* (size-adjusted)	264	0.95	<0.001
Copper	all_strains_with_tip	change in a^* at dose 30	259	0.68	<0.001
Copper	R_mucilaginosa_only	baseline a^*	201	0.93	<0.001
Copper	R_mucilaginosa_only	baseline a^* (size-adjusted)	201	0.95	<0.001
Copper	R_mucilaginosa_only	change in a^* at dose 30	201	0.67	<0.001
Iron	all_strains_with_tip	baseline a^*	197	0.51	<0.001
Iron	all_strains_with_tip	baseline a^* (size-adjusted)	197	0.23	0.00
Iron	all_strains_with_tip	change in a^* at dose 30	133	0.93	<0.001
Iron	R_mucilaginosa_only	baseline a^*	167	0.52	<0.001
Iron	R_mucilaginosa_only	baseline a^* (size-adjusted)	167	0.26	0.01
Iron	R_mucilaginosa_only	change in a^* at dose 30	113	0.93	<0.001
Lead	all_strains_with_tip	baseline a^*	254	0.90	<0.001
Lead	all_strains_with_tip	baseline a^* (size-adjusted)	254	0.90	<0.001
Lead	all_strains_with_tip	change in a^* at dose 30	252	0.86	<0.001
Lead	R_mucilaginosa_only	baseline a^*	199	0.90	<0.001
Lead	R_mucilaginosa_only	baseline a^* (size-adjusted)	199	0.90	<0.001
Lead	R_mucilaginosa_only	change in a^* at dose 30	198	0.86	<0.001
Zinc	all_strains_with_tip	baseline a^*	75	0.02	0.85
Zinc	all_strains_with_tip	baseline a^* (size-adjusted)	75	0.51	0.04
Zinc	all_strains_with_tip	change in a^* at dose 10	75	0.95	<0.001
Zinc	R_mucilaginosa_only	baseline a^*	65	0.02	0.89
Zinc	R_mucilaginosa_only	baseline a^* (size-adjusted)	65	0.60	0.02
Zinc	R_mucilaginosa_only	change in a^* at dose 10	65	0.95	<0.001

Table 9. Pagel's lambda (maximum likelihood under Brownian motion on the PHYling FastTree). 254-264 strains with a tree tip in Cr, Cu and Pb; 197 in Fe; 75 in Zn.

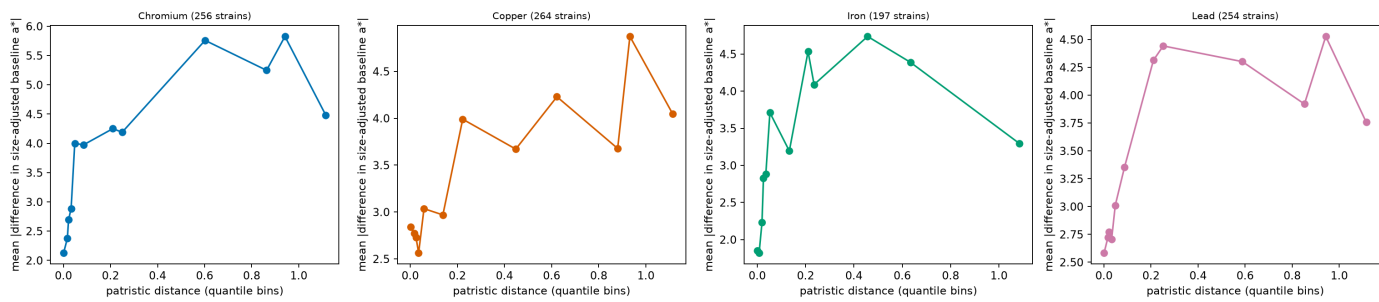


Figure 10. Mean absolute difference in size-adjusted baseline a^* between pairs of strains, by patristic distance.

PHYlog tree (278 tips) with baseline a* (0 dose, strain mean) per metal; colour = a* (viridis, per-metal 2-98th percentile); black = no data

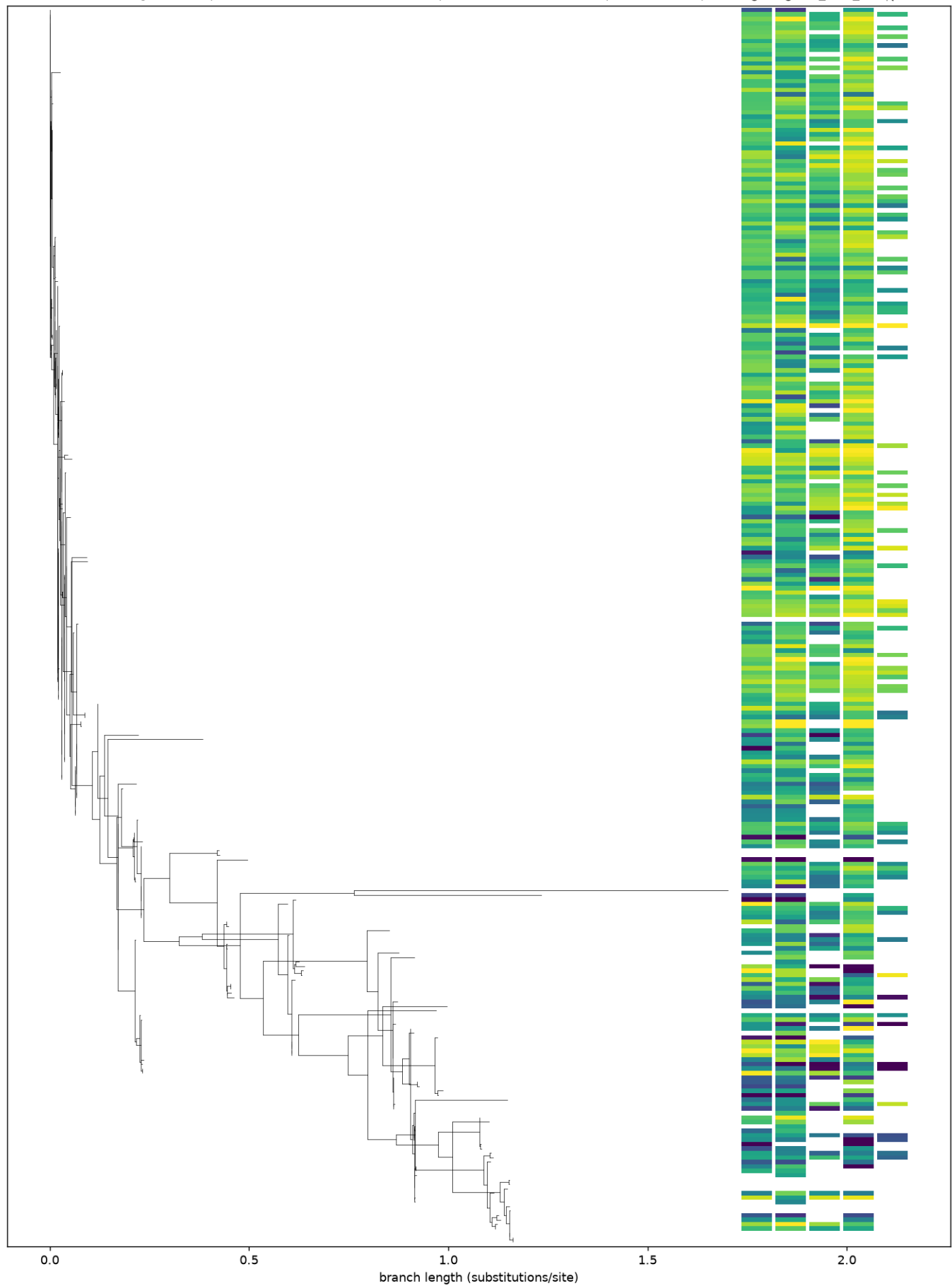


Figure 11. The tree with baseline a^* per metal.

Reading and limits. - Closely related strains share baseline a^* . Lambda is 0.90-0.98 for baseline a^* in Cr, Cu and Pb, also inside *R. mucilaginosa* alone (0.90-0.98), and the pairwise difference in a^* increases with patristic distance (Spearman 0.35 Cr, 0.17 Cu, 0.21 Pb). Fe is weaker (0.51 raw, 0.23 size-adjusted). Baseline colony size has lambda near 0 in Cr, Cu and Pb but 0.83 in Fe and 0.99 in Zn. - **Blomberg's K was dropped.** The tree has 22 zero-length tips and 74 tips with a neighbour closer than $1e-5$, so the covariance matrix is nearly singular (condition number about $1e10$). K varied from $1e-7$ to $4e-2$ with the diagonal jitter (Table 10), so it is not interpretable. Lambda was stable across jitter $1e-8$ to $1e-2$ (0.89 to 0.99 for Cr, Cu and Pb). - Lambda near 1 partly reflects near-identical (clonal) strains having similar a^* . It does not show that a^* evolves under Brownian motion. - 266 of 321 strains have a unique tree tip; strains without a tip are excluded. These are mostly outside the sequenced set.

Metal	jitter	cond	lam	K	pK
Copper	1e-08	1.17e+10	0.95	4.88e-07	0.145
Copper	1e-06	1.17e+08	0.955	1.89e-05	0.19
Copper	0.0001	1.17e+06	0.999	0.000772	0.06
Copper	0.001	1.17e+05	0.999	0.00481	0.005
Copper	0.01	1.17e+04	0.999	0.0309	0.005
Lead	1e-08	9.93e+09	0.894	9.91e-07	0.03
Lead	1e-06	9.93e+07	0.901	3.64e-05	0.06
Lead	0.0001	9.93e+05	0.998	0.00121	0.01
Lead	0.001	9.93e+04	0.998	0.0066	0.005
Lead	0.01	9.93e+03	0.991	0.042	0.005
Chromium	1e-08	1.03e+10	0.965	4.1e-07	0.23
Chromium	1e-06	1.03e+08	0.969	2.06e-05	0.18
Chromium	0.0001	1.03e+06	0.991	0.000855	0.045
Chromium	0.001	1.03e+05	0.975	0.00554	0.005
Chromium	0.01	1.03e+04	0.985	0.0404	0.005

Table 10. Sensitivity of lambda and K to the diagonal jitter added to the covariance matrix.

6. Dose regimes (item 3)

Regimes are defined from colony size: a dose is **sub-inhibitory** when the median (across strains) of area at that dose / area at dose 0 is at least 0.5, and **inhibitory** below that. The 0.5 cutoff is a choice, not a measured threshold.

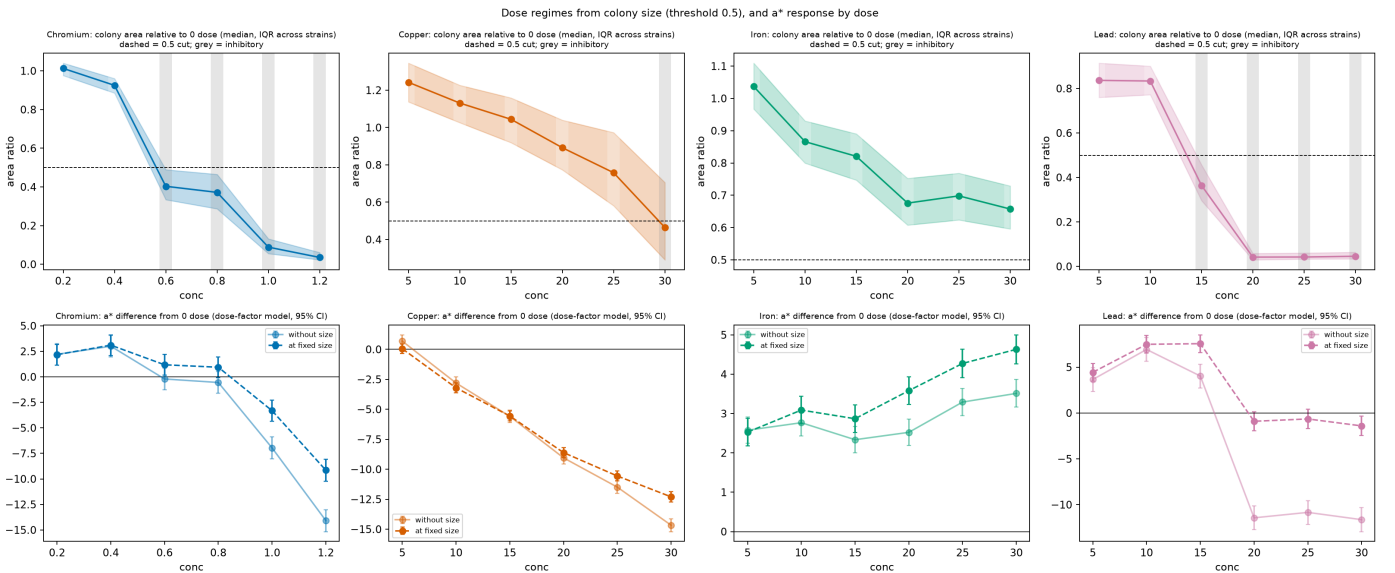


Figure 12. Top: area ratio by dose with the 0.5 cutoff. Bottom: a^* difference from 0 dose, with and without size.

Metal	conc	median_ratio	q25	q75	n_strains	regime
Chromium	0.20	1.01	0.98	1.04	306	sub-inhibitory
Chromium	0.40	0.93	0.89	0.96	306	sub-inhibitory
Chromium	0.60	0.40	0.33	0.49	305	inhibitory
Chromium	0.80	0.37	0.29	0.46	305	inhibitory
Chromium	1.00	0.09	0.06	0.13	301	inhibitory
Chromium	1.20	0.04	0.02	0.06	305	inhibitory
Copper	5.00	1.24	1.14	1.35	319	sub-inhibitory
Copper	10.00	1.13	1.03	1.23	318	sub-inhibitory
Copper	15.00	1.04	0.92	1.16	318	sub-inhibitory
Copper	20.00	0.89	0.77	1.04	317	sub-inhibitory
Copper	25.00	0.76	0.58	0.97	315	sub-inhibitory
Copper	30.00	0.46	0.29	0.71	314	inhibitory
Iron	5.00	1.04	0.97	1.11	219	sub-inhibitory
Iron	10.00	0.87	0.80	0.93	218	sub-inhibitory
Iron	15.00	0.82	0.75	0.89	218	sub-inhibitory
Iron	20.00	0.68	0.61	0.75	218	sub-inhibitory
Iron	25.00	0.70	0.62	0.77	147	sub-inhibitory
Iron	30.00	0.66	0.60	0.73	146	sub-inhibitory
Lead	5.00	0.84	0.76	0.91	303	sub-inhibitory
Lead	10.00	0.83	0.77	0.90	302	sub-inhibitory
Lead	15.00	0.36	0.30	0.46	304	inhibitory
Lead	20.00	0.04	0.03	0.06	301	inhibitory
Lead	25.00	0.04	0.03	0.06	302	inhibitory
Lead	30.00	0.05	0.03	0.06	302	inhibitory

Table 11. Colony area relative to 0 dose.

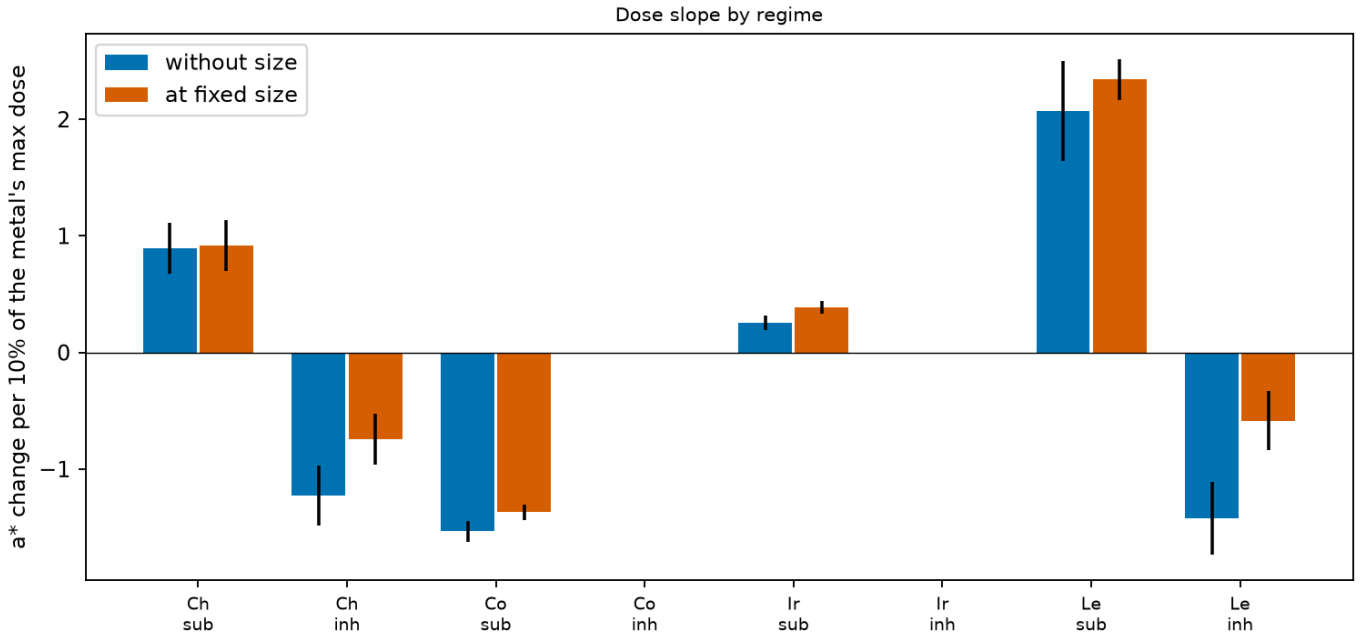


Figure 13. Slope of a^* per 10% of the maximum dose, by regime.

Metal	regime	doses	n_wells	size_adjusted	slope_per_0.1_of_max_dose	p
Chromium	sub-inhibitory	0,0.2,0.4	3577	False	0.89	0.11
Chromium	sub-inhibitory	0,0.2,0.4	3577	True	0.92	0.11
Chromium	inhibitory	0,0.6,0.8,1,1.2	5728	False	-1.23	0.13
Chromium	inhibitory	0,0.6,0.8,1,1.2	5728	True	-0.74	0.11

Metal	regime	doses	n_wells	size_adjusted	slope_per_0.1_of_max_dose	p
Copper	sub-inhibitory	0,5,10,15,20,25	6614	False	-1.53	0.05 <0.001
Copper	sub-inhibitory	0,5,10,15,20,25	6614	True	-1.37	0.03 <0.001
Iron	sub-inhibitory	0,5,10,15,20,25,30	4073	False	0.25	0.03 <0.001
Iron	sub-inhibitory	0,5,10,15,20,25,30	4073	True	0.39	0.03 <0.001
Lead	sub-inhibitory	0,5,10	3061	False	2.07	0.22 <0.001
Lead	sub-inhibitory	0,5,10	3061	True	2.34	0.09 <0.001
Lead	inhibitory	0,15,20,25,30	4717	False	-1.42	0.16 <0.001
Lead	inhibitory	0,15,20,25,30	4717	True	-0.59	0.13 <0.001

Table 12. Regime-specific slopes (a^* change per 10% of the metal's maximum dose; random effects for strain, run and plate).

Reading. - Cr: sub-inhibitory doses (0-0.4) raise a^* by +0.9 per 10% of the range; inhibitory doses lower it (-0.7 at fixed size, -1.2 without size). - Pb: sub-inhibitory doses (0-10) raise a^* by +2.1 to +2.3 per 10%; inhibitory doses (15-30) lower it (-0.6 at fixed size, -1.4 without size). Pb colonies at doses 20-30 are about 4% of control area, so the inhibitory-regime estimate rests on very small colonies. - Cu: no inhibitory regime except the top dose, yet a^* falls steadily (-1.4 per 10% at fixed size): Cu lowers a^* without reducing colony size. - Fe: area never falls below 0.66 of control, and a^* rises slightly (+0.4 per 10% at fixed size).

7. Size-matched comparison (item 4)

Colonies are placed in five size bins (quantiles of $\ln$ area within each metal) and the dose effect is estimated inside each bin.

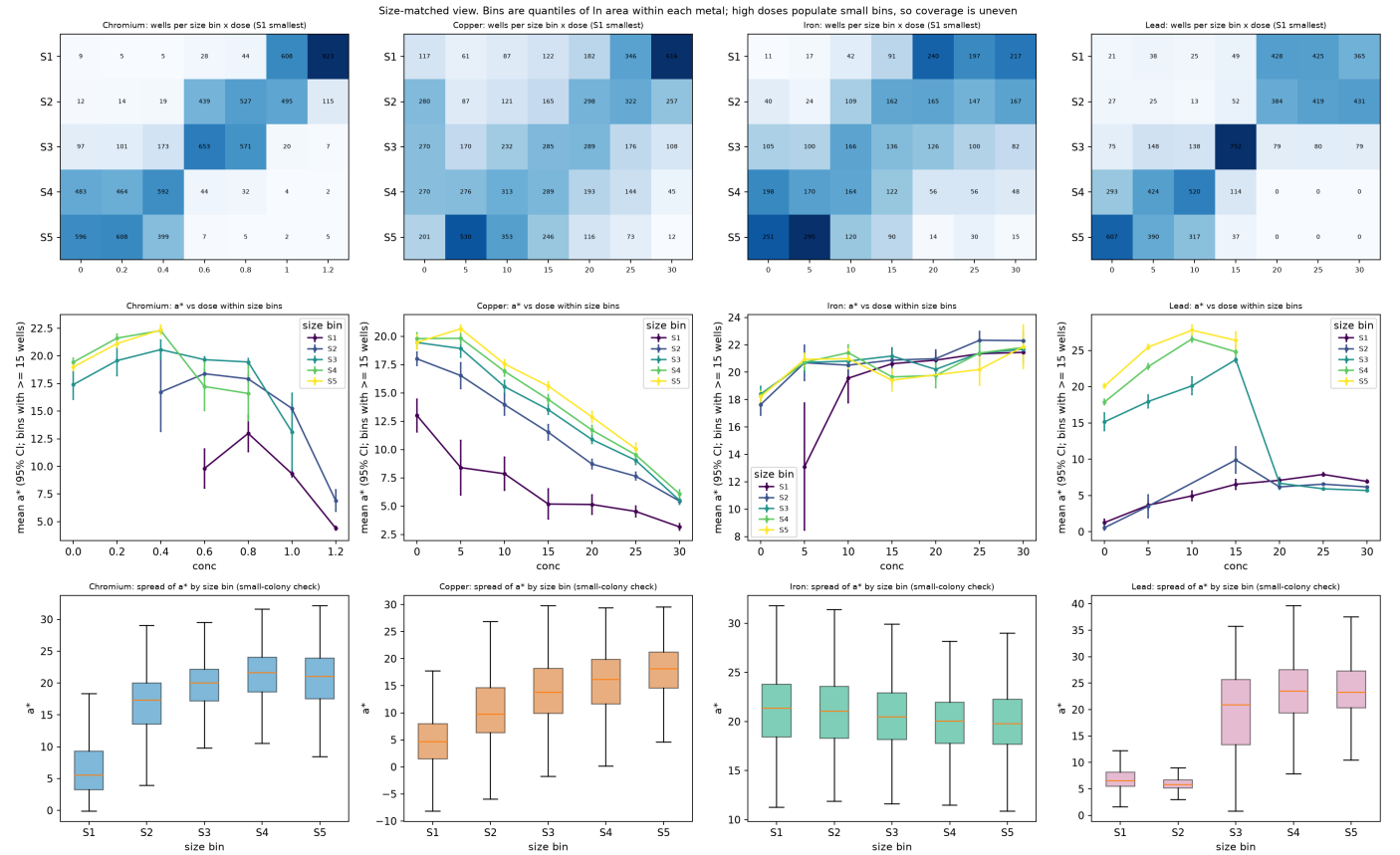


Figure 14. Wells per size bin and dose (top), a^* against dose within bins (middle), and a^* spread by size bin (bottom).

Metal	size_bin	n_wells	n_doses	mean_a	sd_a	slope_full_range	se	p
Chromium	S1	1622	7	6.66	4.45	-5.07	2.03	0.01

Metal	size_bin	n_wells	n_doses	mean_a	sd_a	slope_full_range	se	p
Chromium	S2	1621	7	16.36	5.16	-10.43	1.54	<0.001
Chromium	S3	1622	7	19.38	4.89	-5.35	1.29	<0.001
Chromium	S4	1621	7	20.92	4.85	-8.40	1.46	<0.001
Chromium	S5	1622	7	20.50	4.72	-12.07	2.01	<0.001
Copper	S1	1531	7	5.10	6.15	-10.16	0.62	<0.001
Copper	S2	1530	7	10.81	6.09	-13.52	0.35	<0.001
Copper	S3	1530	7	13.90	5.74	-14.22	0.35	<0.001
Copper	S4	1530	7	15.81	5.39	-14.10	0.44	<0.001
Copper	S5	1531	7	17.78	4.81	-13.62	0.58	<0.001
Iron	S1	815	7	20.71	5.34	3.53	0.68	<0.001
Iron	S2	814	7	21.24	4.17	4.52	0.37	<0.001
Iron	S3	815	7	20.62	3.89	4.20	0.39	<0.001
Iron	S4	814	7	20.17	3.68	4.03	0.40	<0.001
Iron	S5	815	7	19.91	3.90	3.14	0.49	<0.001
Lead	S1	1351	7	7.06	2.61	4.71	0.53	<0.001
Lead	S2	1351	7	6.25	2.58	4.56	0.90	<0.001
Lead	S3	1351	7	19.15	8.37	-14.52	1.68	<0.001
Lead	S4	1351	4	23.36	6.01	17.02	1.35	<0.001
Lead	S5	1351	4	23.63	5.73	19.75	1.23	<0.001

Table 13. Dose effect within size bins (a^* change over the full dose range; random effects for strain, run, plate).

Reading. - Cu: a^* falls with dose in every size bin (-10 to -14), and all seven doses are present in every bin. This is the cleanest size-matched result: Cu lowers a^* at the same size. - Fe: a^* rises in every bin (+3.1 to +4.5). - Cr: negative in all bins (-5 to -12). - Pb: size and dose are strongly confounded. The two smallest bins (below about 1,850 px) have a^* of 6.3-7.1 and SD 2.6 at all doses, and the two largest bins contain only doses 0-15. Slopes in the Pb bins differ in sign (+4.7 in the small bins, -14.5 in S3, +17 to +20 in the large bins), so there is no single size-matched Pb estimate. - **Coverage is uneven.** High doses populate the small bins; in Cr and Pb some size-by-dose cells contain fewer than 10 wells.

8. Selection on baseline, split-half (item 5)

Strains with at least 2 replicate wells at dose 0 and at the top dose. Replicate wells are split at random into halves A and B (1,000 splits). **Naive:** select the top and bottom baseline thirds from all wells and measure the change in the same wells. **Split-half:** select on half A, measure the change in half B.

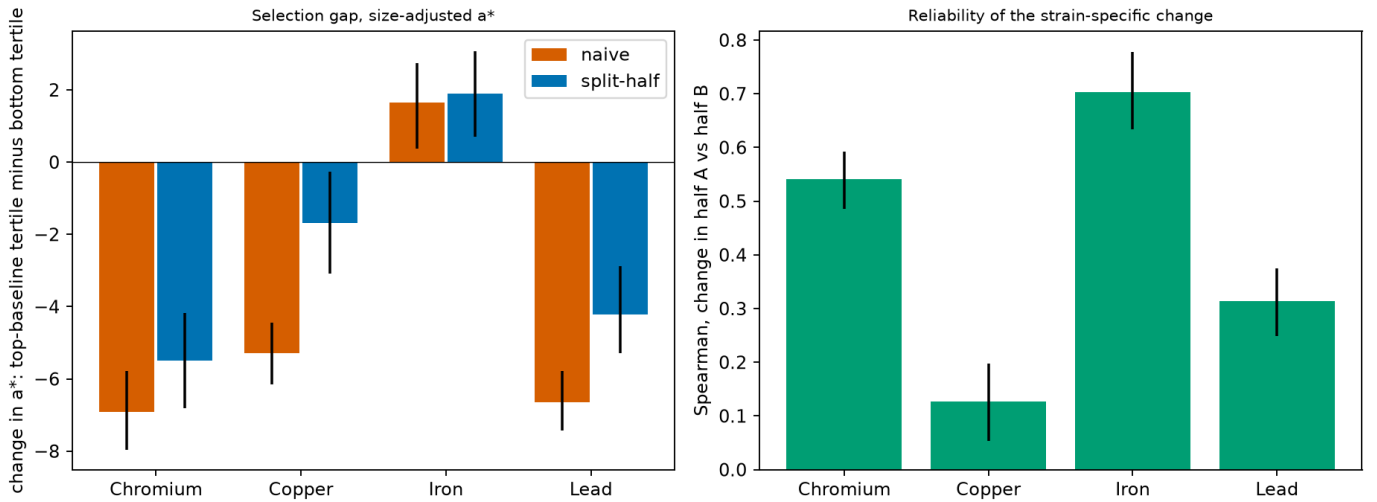


Figure 15. Gap in the change in a^* between the high- and low-baseline thirds (left) and reliability of the strain-specific change (right).

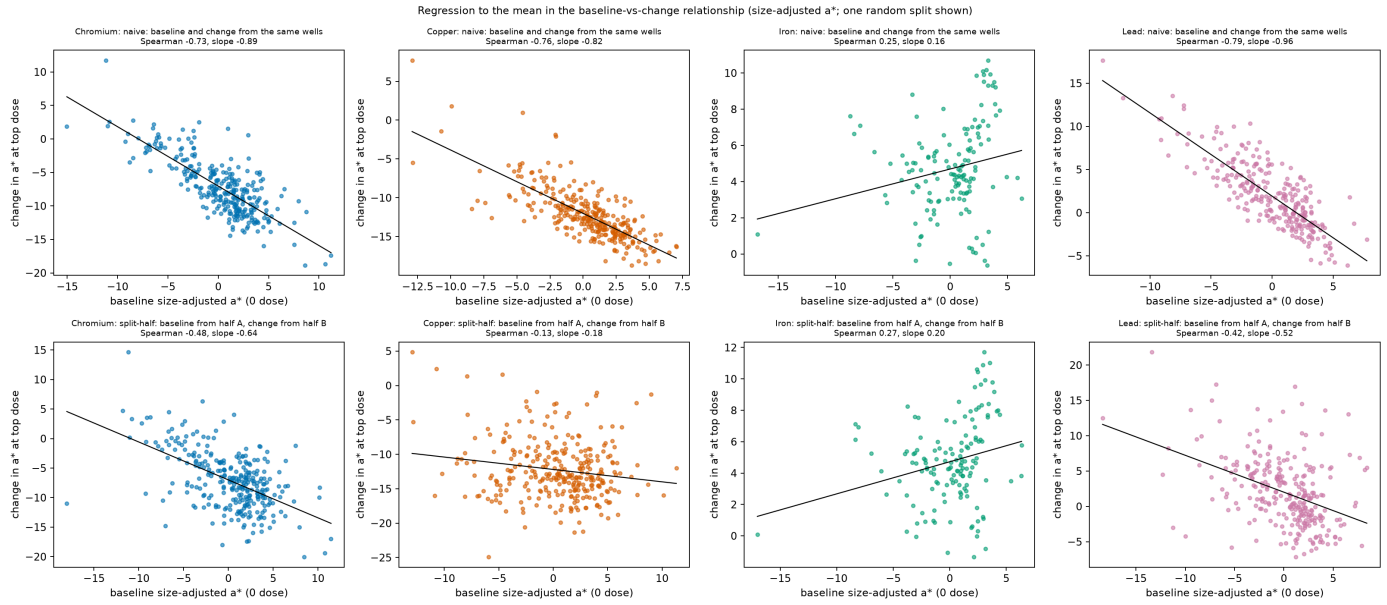


Figure 16. Baseline against change in a^* : same wells (top) and independent halves (bottom).

Metal	top_dose	naive_gap	naive_ci	naive_ci	split_gap	split_gap	split_gap	reliability_change	reliability_B	baseline_A_vs_B
Chromium	20	289	-6.91	-7.96	-5.78	-5.48	-6.81	-4.17	0.54	0.72
Copper	30.00	303	-5.29	-6.16	-4.45	-1.68	-3.08	-0.27	0.13	0.25
Iron	30.00	141	1.64	0.38	2.74	1.89	0.70	3.07	0.70	0.86
Lead	30.00	273	-6.65	-7.42	-5.79	-4.21	-5.28	-2.87	0.31	0.57

Table 14. Size-adjusted a^* . **naive_gap** and **split_gap_mean** are the change in the top-baseline third minus the bottom third (95% CI from a bootstrap over strains). **reliability_change_A_vs_B** is the Spearman correlation of the strain-specific change between the two halves.

Reading. - Part of the strong negative baseline-vs-change relationship in Cr, Cu and Pb is regression to the mean: the gap shrinks from -5.3 to -1.7 in Cu, from -6.9 to -5.5 in Cr and from -6.7 to -4.2 in Pb. Fe is unchanged (+1.6 naive, +1.9 split). - In Cu the strain-specific change is barely repeatable (0.13), and baseline a^* in Cu replicate wells is also noisy (0.25 between halves). - The reliabilities are for halves of 1-2 wells and understate full-data reliability.

9. Batch (item 6)

Each metal is analysed separately, with run and plate as random effects (all models above). Each run is a different set of about 70-84 strains across all doses, so run and strain set are confounded. Zinc is excluded.

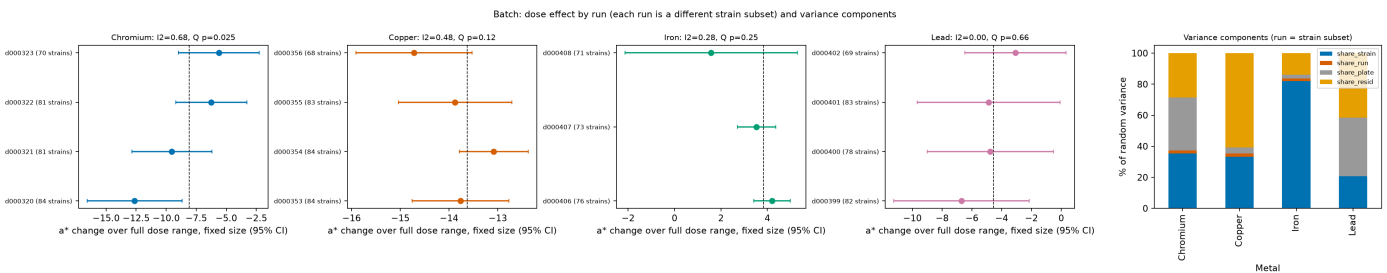


Figure 17. Dose effect at fixed size by run (left four panels) and variance components (right).

Metal	k_runs	pooled_effect	Q	Q_p	I2
Chromium	4	-8.07	9.35	0.03	0.68
Copper	4	-13.63	5.77	0.12	0.48
Iron	3	3.83	2.76	0.25	0.28

Metal	k_runs	pooled_effect	Q	Q_p	I2
Lead	4	-4.55	1.61	0.66	0.00

Table 15. Heterogeneity of the dose effect across runs (I2, Cochran Q).

Metal	run	n_wells	n_strains	n_doses	dose_effect	se	p
Chromium	d000320	2254	84	7	-12.62	2.01	<0.001
Chromium	d000321	2086	81	7	-9.52	1.69	<0.001
Chromium	d000322	2104	81	7	-6.24	1.51	<0.001
Chromium	d000323	1664	70	7	-5.62	1.72	0.00
Copper	d000353	2127	84	7	-13.77	0.51	<0.001
Copper	d000354	1976	84	7	-13.08	0.36	<0.001
Copper	d000355	1920	83	7	-13.87	0.60	<0.001
Copper	d000356	1629	68	7	-14.72	0.61	<0.001
Iron	d000406	1950	76	7	4.20	0.40	<0.001
Iron	d000407	1768	73	7	3.54	0.42	<0.001
Iron	d000408	355	71	5	1.58	1.89	0.46
Lead	d000399	2017	82	7	-6.69	2.32	0.01
Lead	d000400	1663	78	7	-4.76	2.17	0.04
Lead	d000401	1605	83	7	-4.87	2.44	0.06
Lead	d000402	1470	69	7	-3.07	1.73	0.09

Table 16. Dose effect (a^* change over the full dose range at fixed size) per run.

Metal	n_runs	share_strain	share_run	share_plate	share_resid	singular
Chromium	4	0.35	0.02	0.34	0.28	False
Copper	4	0.33	0.02	0.04	0.61	False
Iron	3	0.82	0.02	0.03	0.14	False
Lead	4	0.21	0.00	0.38	0.42	True

Table 17. Variance components (shares of random variance).

Reading. Run explains 0-2% of the random variance. Plate explains 34% (Cr) and 38% (Pb) but only 3-4% in Cu and Fe. The Cr dose effect shrinks monotonically from run d000320 to d000323 (-12.6, -9.5, -6.2, -5.6); this could be a run-order effect or a difference between strain subsets.

10. Minimum colony area (small-colony check)

Very small colonies may have a^* dominated by background pixels. Evidence in the data: in Pb, wells below about 1,850 px have a^* of 6-7 with SD 2-2.7 at every dose (a hinge fit of a^* on $\ln$ area finds a flat segment below 1,846 px, with 30% of Pb wells below it); in Cr, wells below about 1,100-1,800 px have a^* of 4-6 with SD under 3.6. Cu, Fe and Zn colonies are rarely that small (5th percentile of area: Cu 6,759 px, Fe 13,829 px).

To test how much this matters, well-images whose largest object is smaller than a minimum area (1,000, 2,000, 3,000 or 5,000 px) were dropped, and the dose models (strain, run and plate random effects, at fixed size) were refitted. 90%, 78%, 70% and 59% of well-images remain. The 2,000 px value is the main sensitivity setting; the others show how much the result depends on that choice. Zinc is excluded.

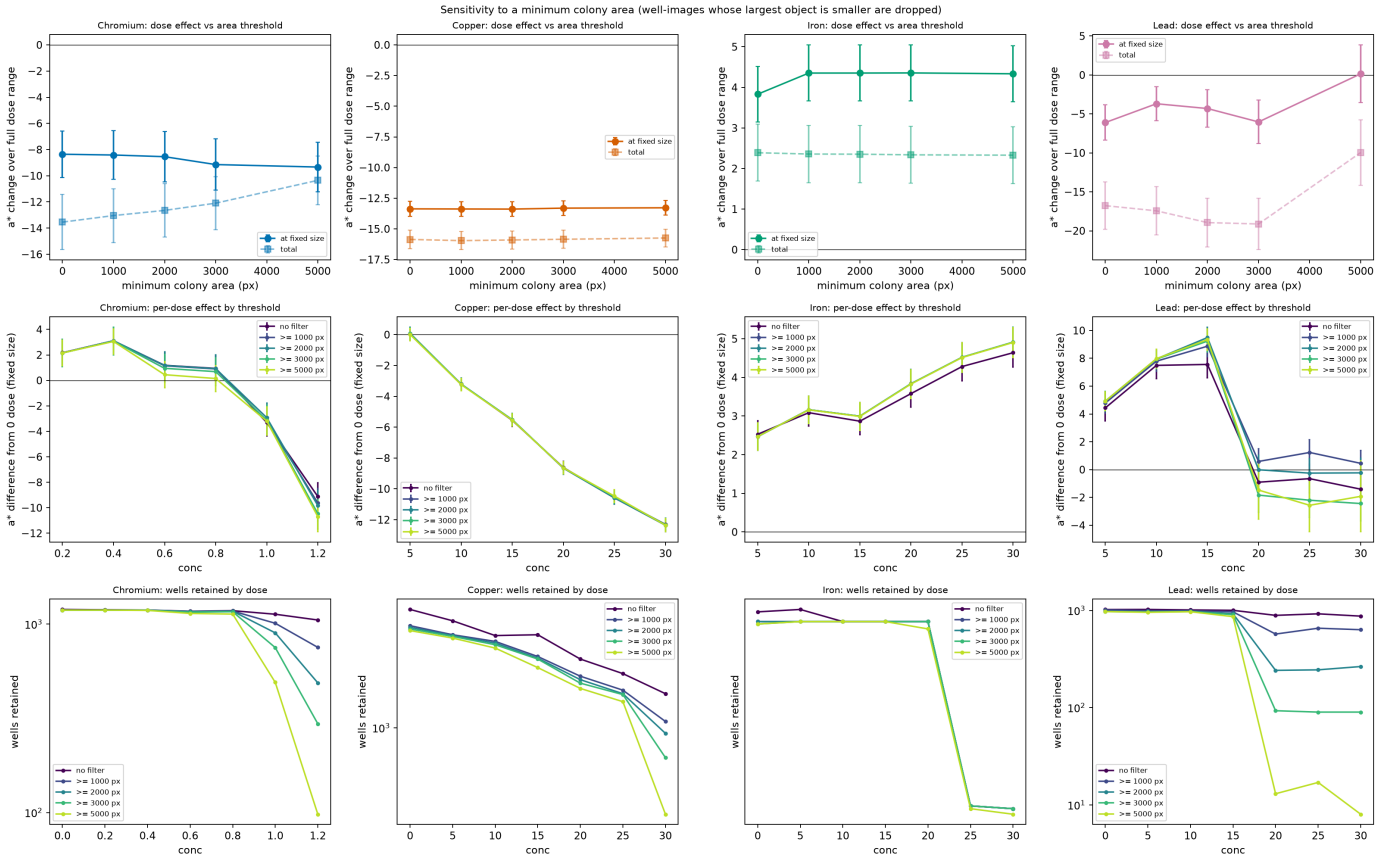


Figure 18. Dose effect against area threshold (top), per-dose effect at fixed size by threshold (middle), and wells retained by dose (bottom).

Metal	0	1000	2000	3000	5000
Chromium	-8.36	-8.41	-8.54	-9.14	-9.33
Copper	-13.37	-13.38	-13.39	-13.31	-13.28
Iron	3.83	4.35	4.35	4.36	4.34
Lead	-6.10	-3.69	-4.31	-6.03	0.15

Table 18. Dose effect on a^* over the full dose range, at fixed size, by minimum colony area (0 = no filter).

dose	effect @0	effect @1000	effect @2000	effect @3000	effect @5000	wells @0	wells @1000	wells @2000	wells @3000	wells @5000
0						1197	1189	1188	1188	1187
0.2	2.2	2.2	2.2	2.2	2.2	1192	1190	1188	1187	1187
0.4	3.1	3.1	3.1	3.1	3.1	1188	1187	1186	1185	1183
0.6	1.2	1.1	1.2	0.9	0.4	1171	1168	1166	1158	1140
0.8	0.9	0.9	0.9	0.7	0.1	1179	1174	1172	1164	1131
1	-3.3	-3.2	-2.9	-3.1	-3.2	1129	1011	899	751	491
1.2	-9.1	-9.6	-9.8	-10.5	-10.7	1052	753	488	296	98

Table 19. Chromium: a^* difference from 0 dose at fixed size, by threshold, and wells retained.

dose	effect @0	effect @1000	effect @2000	effect @3000	effect @5000	wells @0	wells @1000	wells @2000	wells @3000	wells @5000
0						1023	1015	978	975	974
5	4.4	4.8	4.9	4.9	4.9	1025	997	972	962	958

dose	effect @0	effect @1000	effect @2000	effect @3000	effect @5000	wells @0	wells @1000	wells @2000	wells @3000	wells @5000
10	7.5	7.8	8	7.9	8	1013	994	979	975	971
15	7.6	8.9	9.5	9.2	9.3	1004	968	925	906	864
20	-0.9	0.6	-0	-1.8	-1.5	891	572	242	93	13
25	-0.6	1.2	-0.3	-2.2	-2.6	924	657	245	90	17
30	-1.4	0.4	-0.2	-2.4	-1.9	875	634	265	90	8

Table 20. Lead: a^* difference from 0 dose at fixed size, by threshold, and wells retained.

Metal	min_area	top_dose	n_strains_top	top_dose_mean	atop_dose_sd	across_strains	median_area_top_px
Chromium	0	1.20	311	4.43	2.52		1728.99
Copper	0	30.00	314	4.06	2.64		16541.21
Iron	0	30.00	146	21.64	4.38		19508.39
Lead	0	30.00	307	6.48	1.09		1429.57
Chromium	1000	1.20	284	4.95	3.04		2405.72
Copper	1000	30.00	309	4.34	2.38		16974.50
Iron	1000	30.00	146	21.64	4.38		19508.39
Lead	1000	30.00	291	6.32	1.17		1794.31
Chromium	2000	1.20	241	5.12	3.57		3281.38
Copper	2000	30.00	308	4.45	2.35		17287.13
Iron	2000	30.00	146	21.64	4.38		19508.39
Lead	2000	30.00	189	5.87	0.77		2712.98
Chromium	3000	1.20	177	5.53	4.11		4101.66
Copper	3000	30.00	307	4.52	2.34		17960.48
Iron	3000	30.00	146	21.64	4.38		19508.39
Lead	3000	30.00	81	5.67	0.56		3476.00
Chromium	5000	1.20	74	7.13	6.15		6516.56
Copper	5000	30.00	306	4.66	2.37		19181.04
Iron	5000	30.00	145	21.81	3.87		19522.91
Lead	5000	30.00	8	5.28	0.98		6610.00

Table 21. Strain-mean a^* at the top dose by threshold.

Reading. - **Cu and Fe are unaffected** (Cu -13.4 to -13.3; Fe +3.8 to +4.4). Their colonies are large at every dose. - **Cr: the top-dose drop is not a small-colony artifact.** The effect at dose 1.2 is -9.1 with no filter and -10.7 with colonies of at least 5,000 px (98 wells, 74 strains remain). The effects at doses 0.2-0.8 do not change. The overall dose effect stays between -8.4 and -9.3. - **Pb: the rise at doses 5-15 holds or strengthens** with a threshold (+4.9, +8.0 and +9.5 at doses 5, 10 and 15 at 2,000 px, against +4.4, +7.5 and +7.6 with no filter). At doses 20-30 only 242-265 wells remain at 2,000 px, 90 at 3,000 px and 8-17 at 5,000 px (572-657 at 1,000 px), so the inhibitory range of Pb cannot be tested. In those few colonies the fixed-size difference from control is between -2.6 and +1.2. The Pb overall dose effect is unstable across thresholds (-6.1, -3.7, -4.3 [singular fit], -6.0, +0.2) because it depends on how many top-dose wells are left, and should not be quoted. - **A floor remains in Pb.** Among Pb colonies of at least 3,000 px at dose 30 (81 strains), strain-mean a^* is still 5.7 with SD 0.56 across strains. Area thresholds alone do not tell us whether this is background or a real loss of pigment. - Fixed-size effects extrapolate a size- a^* slope fitted mostly on larger colonies; thresholds keep the comparison within the range where colonies of similar size exist, which is why the Pb inhibitory doses lose almost all their wells.

Limits

- Dose units are not given in the source, and metals are not pooled.
- Zinc is not modelled in the stratified analyses (one plate per dose).
- Size is partly an effect of stress; models “at fixed size” estimate a direct effect, not the whole induction. One size slope is used for all strains.
- Survivor selection: wells that never produced objects are absent, and wells with fewer than 2 window images were dropped (see `report/tables/wells_lost_by_dose.csv`).

- 16 strains still have no species and are excluded from species tests (see the data-problems report). Species tests use species with at least 5 strains; *R. mucilaginosa* has 174-216 strains and the others have 5-18.
- Population labels come from an earlier GWAS run (201 strains).
- The all-objects sensitivity table is built (**results/wells_allobj.csv**) but not modelled.
- Old-vs-new strain assignment for Copper is unresolved (see the data-problems report). This report uses the new data's own strain labels.